Homework Exercises 8: Band-structure of Polyacetylene

Condensed Matter Theory I

Thursday 25th February, 2021

In this exercise we will investigate the band-structure of Polyacetylene $(C_2H_2)_\infty$. We will consider trans-$(C_2H_2)_\infty$ where the C atoms form a zigzag chain in the xy-plane with two C atoms in one unit cell. The positions of the Carbon atomis are approximately $\{2na, (1/2)\sqrt(1/3)a, 0\}$ and $\{(2n+1)a, -(1/2)\sqrt(1/3)a, 0\}$, with n an integer.

In order to understand its electronic structure one needs to consider the C 2s and 2p orbitals and the H 1s orbital. The C $2s$, $2p_x$ and $2p_y$ orbitals form an sp_2 hybrid orbital `https://en.wikipedia.org/wiki/Orbital_hybridisation#sp2`, wich results in a planar structure where each C atom has 3 neighbours bonding to the 3 lobes of the 3 possible sp_2 orbitals. One of the bonds is with the H atom, with 2 electrons in the bonding orbital and the anti-boding orbital empty. The other two lobes of the sp_2 orbital point to neighbour C atoms. Again in this case the bonding orbitals are filled, the anti-bonding orbitals are empty.

Each CH unit thus has 4 electrons in bonding orbitals (2 internal for the C-H bond, 1 with each neighbour) which leads to a stable structure. Electron counting (C provides 4 electrons H 1) leads to the conclusion there is one electron left. This last electron occupies the p_z orbital that is perpendicular to the sp_2 orbitals. This p_z orbital makes a π bond to the neighbour C atom.

1 Equal spacing

1. Write down the tight-binding Hamiltonian of the p_z orbital in polyacetylene. Assume only nearest neighbour interaction, take two C atoms per unit cell of length $2a$ but assume equidistant seperation between the C atoms.

$$H = \sum_{i=-\infty}^{\infty} \epsilon_p (a_i^\dagger a_i + b_i^\dagger b_i) + pp\pi (a_i^\dagger b_i + b_i^\dagger a_i + b_i^\dagger a_{i+1} + a_{i+1}^\dagger b_i) \tag{1}$$

   Here a_i is the annihilation operator associated with the left of the two atoms in the unit cell while b_i is the annihilation operator for the right one.

2. Show that you can simplify this tight binding Hamiltonian by reducing the unit cell to effectively only contain one C atom. Use this smaller unit cell to answer the following questions:
   Since in the above Hamiltonian in eq. 1 there is no difference in onsite energy or

hopping between the atoms a and b we can just use a smaller unit cell to get the following Hamiltonian:

$$H = \sum_{i=-\infty}^{\infty} \epsilon_p (a_i^\dagger a_i) + pp\pi (a_i^\dagger a_{i+1} + a_{i+1}^\dagger a_i) \tag{2}$$

(a) What is the size of the Brillouin zone in that case?
$\vec{a}\vec{b} = 2\pi \implies \vec{b} = 2\pi/a$ where $\vec{a}$ is the lattice vector of length a in one dimension and $\vec{b}$ is the corresponding reciprocal lattice vector. The first Brillouin zone in this case extends from $-\pi/a$ to π/a in k-space.

(b) Calculate and plot the band structure of polyacetylene.
We diagonalize by using the Fourier transform:

$$\begin{aligned} a_j^\dagger &= \frac{1}{\sqrt{Na}} \sum_k e^{-ik(ja)} a_k^\dagger, \\ a_j &= \frac{1}{\sqrt{Na}} \sum_k e^{ik(ja)} a_k. \end{aligned} \tag{3}$$

By plugging the above equation into the Hamiltonian and exploiting the identity $\frac{1}{N}\sum_{n=1}^{N} e^{-i(k-k')na} = \delta_{k,k'}$ we arrive at the diagonalized form of the Hamiltonian:

$$H = \sum_k (\epsilon_p + 2pp\pi \cos(ka)) a_k^\dagger a_k \tag{4}$$

from which we can read of the band structure:

$$\epsilon(k) = \epsilon_p + 2pp\pi \cos(ka). \tag{5}$$

(c) Compute the Green's function in k-space (Crystal-momentum) and plot the poles of the imaginary part as a function of momentum.

$$\begin{aligned} G(k,k',\omega) &= \langle 0| a_k \frac{1}{\omega + i0^+ - H(k)} a_k^\dagger |0\rangle = \langle 0| a_k \frac{1}{\omega + i0^+ - \epsilon_p - 2pp\pi\cos(ka)} a_k^\dagger |0\rangle \\ &= \delta_{k,k'} \frac{1}{\omega + i0^+ - \epsilon_p - 2pp\pi\cos(ka)} \end{aligned} \tag{6}$$

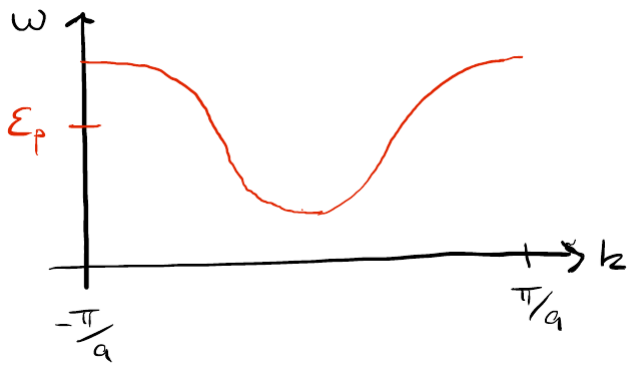
To identify the poles in the imaginary part of the Greens function we make use of the Dirac formula:

$$\lim_{\eta \to 0^+} \frac{1}{E + i\eta} = \mathcal{P}(\frac{1}{E}) - i\pi\delta(E), \tag{7}$$

Where $\mathcal{P}$ signifies the principal value of $\frac{1}{E}$. Therefore we have poles in the imaginary part whenever

$$\omega = \epsilon_p + 2pp\pi\cos(ka), \tag{8}$$

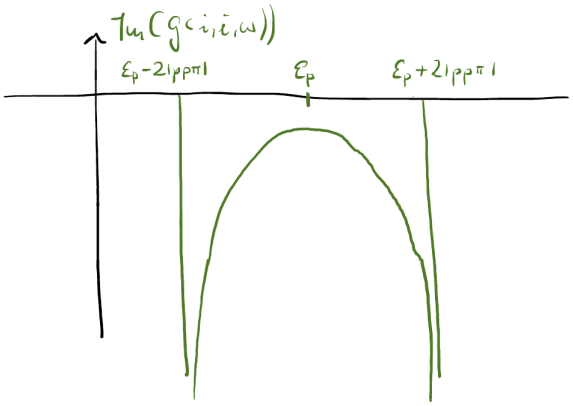
which is exactly the band structure (dispersion relation) as expected.



(d) Compute the local Green's function in real space (Wannier functions) and plot the the imaginary and real part as a function of energy.
The local Greens function is:

$$\begin{aligned}
G(i,i,\omega) = \langle 0|\, a_i \frac{1}{\omega + i0^+ - H} a_i^\dagger\, |0\rangle &= \frac{1}{N}\sum_{k,k'} e^{i(k-k')R_i} G(k,k',\omega) \\
&= \frac{1}{N}\sum_{k,k'} e^{i(k-k')R_i}\delta_{k,k'} \frac{1}{\omega + i0^+ - \epsilon_p - 2pp\pi\cos(ka)} \\
&= \frac{1}{N}\sum_{k} \frac{1}{\omega + i0^+ - \epsilon_p - 2pp\pi\cos(ka)} (\frac{2\pi}{L})(\frac{L}{2\pi}) \\
&\approx \frac{a}{2\pi}\int_{-\pi/a}^{\pi/a} dk \frac{1}{\omega + i0^+ - \epsilon_p - 2pp\pi\cos(ka)} \\
&= \frac{1}{2\pi}\int_{-\pi}^{\pi} dk' \frac{1}{\omega + i0^+ - \epsilon_p - 2pp\pi\cos(k')} \\
&= \sqrt{\frac{1}{(\omega - \epsilon_p + \mathrm{i}0^+ + 2pp\pi)(\omega - \epsilon_p + \mathrm{i}0^+ - 2pp\pi)}}
\end{aligned} \tag{9}$$

Which is imaginary for $(\omega - \epsilon_p)^2 < 4pp\pi^2$.



(e) Calculate and plot the density of states of polyacetylene.
The density of states is:

$$DOS(\epsilon) = -\frac{1}{\pi}\Im(G(i,i,\omega)) \tag{10}$$

with the local Greensfunction as calculated above. The density of states thus has the form of the imaginary part if $G(i,i,\omega)$ as seen in the scetch above.

You can use the following integral:

$$1/(2\pi)\int_{-\pi}^{\pi}\frac{1}{\omega+\mathrm{i}\nu+2\cos[k]}dk = \sqrt{\frac{1}{(\omega+\mathrm{i}\nu+2)(\omega+\mathrm{i}\nu-2)}} \tag{11}$$

2 Peierls dimerization

In the previous problem we described Polyacetylene by one C atom per unit cell. In this problem we will use the proper unit cell of two C atoms per unit cell.

1. With two atoms per unit cell:

   (a) What is the size of the Brillouin zone in the case of two atoms per unit cell
   As the unit cell is now twice as large and the Brillouin zone becomes half as large and extends from $-\pi/2a$ to $\pi/2a$ in k-space.

   (b) Compute the Green's function in k-space and represent it as a two by two matrix related to the two atoms in the unit cell.
   This is very similar to what we have already done on exercise sheet 6. We Fourier transform the components a and b seperately:

$$a_j^\dagger = \frac{1}{\sqrt{Na/2}}\sum_k e^{-ik(2j)a}a_k^\dagger \text{ and } b_j^\dagger = \frac{1}{\sqrt{Na/2}}\sum_k e^{-ik(2j+1)a}b_k^\dagger. \tag{12}$$

   which in just the same way as on exercise sheet 6 results in

$$H = \sum_k \epsilon_p(a_k^\dagger a_k + b_k^\dagger b_k) + 2pp\pi\cos(ka)(a_k^\dagger b_k + b_k^\dagger a_k) \tag{13}$$

which can be expressed in matrix form as:

$$H = \sum_k \begin{pmatrix} a_k^\dagger \\ b_k^\dagger \end{pmatrix}^T \begin{pmatrix} \epsilon_p & 2pp\pi\cos(ka) \\ 2pp\pi\cos(ka) & \epsilon_p \end{pmatrix} \begin{pmatrix} a_k \\ b_k \end{pmatrix}. \tag{14}$$

We see that the Hamiltonian is Block-diagonal in k-space. Therefore we can, for every k-value, calculate 4-nonzero Greens functions that start with $a_k^\dagger |0\rangle$ or $b_k^\dagger |0\rangle$ and end with $\langle 0| a_k$ or $\langle 0| b_k$. These 4 Greensfunctions are simultaneously calculated when we take the Greens operator of H (as H is Block-diagonal in H, the Greens operator is going to be block diagonal as well). One block, corresponding to a k value, is given by:

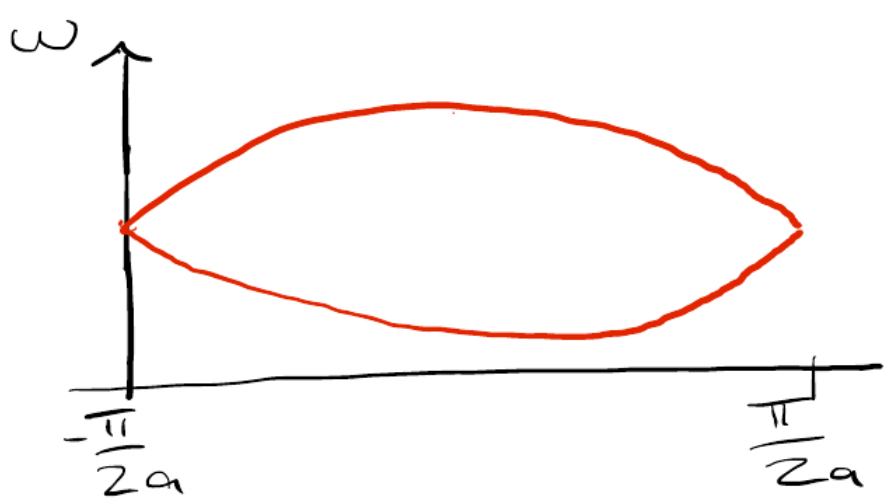
$$\begin{aligned} \hat{G}(k,\omega) =& \frac{1}{(\omega + i0^+) - H_k} = \begin{pmatrix} \omega + i0^+ - \epsilon_p & -2pp\pi\cos(ka) \\ -2pp\pi\cos(ka) & \omega + i0^+ - \epsilon_p \end{pmatrix}^{-1} \\ =& \frac{1}{(\omega - \epsilon_p)^2 - 4pp\pi^2\cos^2(ka)} \begin{pmatrix} \omega + i0^+ - \epsilon_p & 2pp\pi\cos(ka) \\ 2pp\pi\cos(ka) & \omega + i0^+ - \epsilon_p \end{pmatrix} \end{aligned} \tag{15}$$

The 4 entries of this block of the Greens operator are the 4 nonzero Greens functions at a k-value. Thus this can also be considered the Greensfunction at a k-value as a 2 by 2 matrix.

(c) Plot the band-structure as the energy momentum dispersion of the poles of the Green's function.
We get the band structure by diagonalizing the k-blocks in the Hamiltonian which results in

$$H = \sum_k \begin{pmatrix} d_k^\dagger \\ e_k^\dagger \end{pmatrix}^T \begin{pmatrix} \epsilon_p + 2pp\pi\cos(ka) & 0 \\ 0 & \epsilon_p - 2pp\pi\cos(ka) \end{pmatrix} \begin{pmatrix} d_k \\ e_k \end{pmatrix}. \tag{16}$$

Therefore we get the following band structure in the smaller Brillouin zone:



2. Polyacetylene shows a Peirels dimerisation whereby alternating C-C bondlenghts are short and long. Assuming that the change in hopping is δppπ for the shorter bond and -δppπ for the longer bond. Write down H_1 the perturbing Hamiltonian to H of execise (1) that describes this perturbation as a tight binding Hamiltonian in real space (Wannier functions).

$$H_1 = \sum_{i=-\infty}^{\infty} -\delta pp\pi(a_i^\dagger b_i + b_i^\dagger a_i) + \delta pp\pi(b_i^\dagger a_{i+1} + a_{i+1}^\dagger b_i) \tag{17}$$

3. Calculate the Green's function (in k-space as a two by two matrix) of the dimerized structure. Hint: transform H_1 to k-space and use the Dyson equations.
We first Fourier transform the two components a and b in H_1 just as we did with H before resulting in (compare sheet 6):

$$H_1 = \sum_k \begin{pmatrix} a_k^\dagger \\ b_k^\dagger \end{pmatrix}^T \begin{pmatrix} 0 & -2i\delta pp\pi \sin(ka) \\ +2i\delta pp\pi \sin(ka) & 0 \end{pmatrix} \begin{pmatrix} a_k \\ b_k \end{pmatrix}. \tag{18}$$

This is again Block-diagonal in k-space. We could now just calculate the Greens operators of the k-blocks of $H = H_0 + H_1$, where H_0 is the original Hamiltonian from eq. 14. But here we want to make use of the Dyson equation. Therefore we recall that $G_0(k, k', \omega)$ denotes the unperturbed Greens function based on H_0 alone while $G(k, k', \omega)$ is the full Greens function based on H. Also we keep in mind that $G_0(k, k', \omega)$ is now a 2 by 2 matrix that corresponds to starting with a plane wave (with k) of the a or b atoms and ending with a plane wave of the a or b atoms (with k'). We have further already noticed that since H_0 is block-diagonal in k-space so is the Greens operator of which $G_0(k, k', \omega)$ is a submatrix. Therefore $G_0(k, k', \omega)$ is only nonzero if $k = k'$. The last thing we need to keep in mind is that the pertubation Hamiltonian H_1 is also block-diagonal in k-space such that we can only scatter from a k-state to a state with the same k-vector. With these facts in mind the Dyson equation becomes simple:

$$\begin{aligned} G(k, k', \omega) =& G_0(k, k', \omega) + \sum_{l,m} G_0(k, l, \omega) H_1^{l,m} G(m, k', \omega) \\ =& G_0(k, k, \omega) + G_0(k, k, \omega) H_1^{k,k} G(k, k', \omega) \end{aligned} \tag{19}$$

This equation can be solved for $G(k, k', \omega)$ which gives

$$G(k, k', \omega) = \frac{1}{1 - G_0(k, k, \omega) H_1^{k,k}} G_0(k, k, \omega). \tag{20}$$

keep in mind that we are now dealing with matrices and the inverse has to be to the

left! Expressed in matrix form this is

$$
\begin{aligned}
G(k,k',\omega) = &\Bigg[\mathbb{1} - \frac{1}{(\omega-\epsilon_p)^2 - 4pp\pi^2\cos^2(ka)} \begin{pmatrix} \omega + i0^+ - \epsilon_p & 2pp\pi\cos(ka) \\ 2pp\pi\cos(ka) & \omega + i0^+ - \epsilon_p \end{pmatrix} \times \\
&\begin{pmatrix} 0 & -2i\delta pp\pi\sin(ka) \\ +2i\delta pp\pi\sin(ka) & 0 \end{pmatrix}\Bigg]^{-1} \frac{1}{(\omega-\epsilon_p)^2 - 4pp\pi^2\cos^2(ka)} \begin{pmatrix} \omega + i0^+ - \epsilon_p & 2pp\pi\cos(ka) \\ 2pp\pi\cos(ka) & \omega + i0^+ - \epsilon_p \end{pmatrix} \\
= &\frac{1}{\kappa} \begin{pmatrix} \omega + i0^+ - \epsilon_p & -2(pp\pi\cos(ka) - i\delta pp\pi\sin(ka)) \\ -2(pp\pi\cos(ka) + i\delta pp\pi\sin(ka)) & \omega + i0^+ - \epsilon_p \end{pmatrix},
\end{aligned}
\tag{21}
$$

with $\kappa = (\omega-\epsilon_p)^2 - 4(pp\pi^2\cos^2(ka) - i\delta pp\pi^2\sin^2(ka))$. This is the result we would have gotten also if we just calculated the Greens operator for each k-block of $H = H_0 + H_1$.

3 Spectral weight

In the first exercise we calculated the band-structure of Polyacetylene using one atom per unit cell. In the second exersize we calculated the band-structure of Polyacetylene using two atoms per unit cell. Without a distortion the two systems should be the same. The resulting band-structures are however different. With the use of photo electron spectroscopy one can measure the band-structure. We now found two different theoretical answers for a single observable.

In order to solve this apparent contradiction we label the two atoms within the unit cell as C_A and C_B. The Green's function is given as a two by two matrix related to the propagation of creating a particle in atom C_A or C_B and subsequent annihilating it at atom C_A or C_B.

1. Calculate the Green's function if one creates a particle in state $\alpha\psi_A + \beta\psi_B$. Whereby $\alpha^2 + \beta^2 = 1$.

$$
\begin{aligned}
G(\psi,\psi,\omega) =& \langle 0| (\alpha^* a_k + \beta^* b_k)\hat{G}(\alpha a_k^\dagger + \beta b_k^\dagger)|0\rangle \\
=& |\alpha|^2 G(a,a,\omega) + |\beta|^2 G(b,b,\omega) + \alpha^*\beta G(b,a,\omega) + \alpha\beta^* G(a,b,\omega) \\
=& \frac{1}{\kappa}((|\alpha|^2 + |\beta|^2)(\omega-\epsilon_p) - 2pp\pi(\alpha^*\beta + \alpha\beta^*)\cos(ka))
\end{aligned}
\tag{22}
$$

where the different Greensfunctions here are just the components of the matrix in eq. 21, where we again set $\delta pp\pi = 0$.

2. For each k-point this Green's function has two poles. How do the residues of these poles depend on α and β.
For each k-point we have two poles for such a Greensfunction because $\kappa = (\omega-\epsilon_p)^2 - 4(pp\pi^2\cos^2(ka))$ has two roots at $\omega_{1/2} = \epsilon_p \pm 2pp\pi\cos(ka)$. The Residues of these

simple poles at ω_1 and ω_2 are:

$$
\begin{aligned}
Res_1 &= \lim_{\omega\to\omega_1}(\omega-\omega_1)G(\psi,\psi,\omega) \\
&= \frac{1}{\epsilon_p + 2pp\pi\cos(ka) - \epsilon_p + 2pp\pi\cos(ka)}((\epsilon_p + 2pp\pi\cos(ka) - \epsilon_p) - 2pp\pi(\alpha^*\beta + \alpha\beta^*)\cos(ka) \\
&= \frac{1}{4pp\pi\cos(ka)}((2pp\pi\cos(ka)) - 2pp\pi(\alpha^*\beta + \alpha\beta^*)\cos(ka)) \\
Res_2 &= \lim_{\omega\to\omega_2}(\omega-\omega_2)G(\psi,\psi,\omega) \\
&= \frac{1}{\epsilon_p - 2pp\pi\cos(ka) - \epsilon_p - 2pp\pi\cos(ka)}((\epsilon_p - 2pp\pi\cos(ka) - \epsilon_p) - 2pp\pi(\alpha^*\beta + \alpha\beta^*)\cos(ka) \\
&= \frac{1}{-4pp\pi\cos(ka)}((-2pp\pi\cos(ka)) - 2pp\pi(\alpha^*\beta + \alpha\beta^*)\cos(ka))
\end{aligned}
\tag{23}
$$

3. Which value of α and β should one take to retrive a case where one pole has a zero residue and the other a residue of 1? If we choose $\alpha^*\beta + \alpha\beta^* = 2\Re(\alpha^*\beta) = 1$ the residue of the pole at ω_1 vannishes while the residue of the pole at ω_2 becomes 1. The other extreme where the first pole becomes 1 while the other one vannishes is archived by choosing: $\alpha^*\beta + \alpha\beta^* = 2\Re(\alpha^*\beta) = -1$. The superpositions of the k-waves of the a and b atoms such that only one pole has a finite residue correspond to the single poles we have if we consider a smaller unit cell and therfore a larger Brillouin zone.